Investigating the Impact of Generative AI Tutoring Styles on Student Knowledge Acquisition and Confidence Calibration

Kevin Bryan
kevinb@student.unimelb.edu.au

Word count: 1367 (current draft)

[DELETE BEFORE SUBMISSION] Section word-count notes (current / target for 6000 words):
Introduction: 1153 / 1200 Literature Review: 19 / 3000 Research Question: 106 / 500
Research Plan: 10 / 1000 Conclusion: 1 / 300

Department of Computing and Information Systems
The University of Melbourne

March 22, 2026

Contents

1 Introduction **3**

2 Literature Review **4**

2.1 Generative AI in Education 4

2.2 Learning strategies: direct, Socratic, and learning-by-teaching 4

2.3 Confidence calibration and psychological impacts 4

2.4 Research gap 4

3 Research Question **4**

4 Research Plan **4**

4.1 Methodology 4

4.2 Data and experiment 4

4.3 Participants and ethics 4

4.4 Timeline 4

5 Conclusion **4**

1 Introduction

The rise of Generative AI (GenAI) and its capabilities has become a major topic of discussion in the education industry [1]. Ever since the release of ChatGPT by OpenAI in November 2022, the education landscape has evolved tremendously [1]. Whereas institutions initially banned the use of AI, the trend has shifted to embracing and integrating GenAI into pedagogy [2]. This transition is supported by GenAI's ability to deliver personalized learning and customized feedback based on individual student needs [3]. By utilizing GenAI, students are now able to focus on higher-order skills such as critical thinking and creativity, while AI scaffolds and automates repetitive tasks [4, 5]. Moreover, due to today's rapidly evolving professional landscape, the adoption of GenAI in higher education has become a necessity rather than an option [6]. Hence, institutions need to create an environment where AI can be used to enhance human intelligence rather than replace it [1, 5].

However, the adoption of GenAI in higher education comes with a significant set of challenges. As previously mentioned, GenAI can be useful in taking over repetitive tasks [7], but this also leads to "cognitive offloading" where students outsource critical cognitive tasks to the AI [7]. It becomes even more concerning as AI can also create a "productivity paradox," where students experience increased productivity but lack depth in their critical analysis [8]. For instance, due to convenience and speed that AI summaries provide, students tend to prioritize these summaries over verifying the underlying sources [9]. Despite the faster drafting process, the final result ended up being lower quality, with generic essays and a lack of critical analysis [1, 8]. Moreover, this matter is further supported by the so-called "death of essays", a phenomenon that has risen to prominence since COVID-19 due to the increase in take-home exams and students utilizing AI to cheat and plagiarize [1]. This raises the importance of AI literacy. Studies show that students with higher AI literacy tend to take a collaborative approach by planning and evaluating AI outputs, which consequently leads to better outcomes compared to students with lower AI literacy, who tend to over-rely on AI just to get the job done [10]. Additionally, over-reliance on AI can lead to a decline in face-to-face interactions, negatively impacting students' interpersonal skills and emotional intelligence. Thus, increasing the risk of social anxiety and isolation [11].

As previously mentioned, over-reliance on AI is a significant concern that must be addressed. AI tends to generate text with a convincing tone, giving an illusion that the output is always true, causing users to overlook hallucinations [1, 9]. To combat this, students are required to possess foundational knowledge before adopting the tool. This ensures they are able to critically validate the AI's information, as the human in the loop is responsible for the final output produced [1, 12]. Therefore, students must be taught to verify AI outputs [13]. On the other hand, educators are encouraged to create critical thinking assessments that can't be replicated by AI, such as video presentations or pitches [13, 1].

This gap creates a strong opportunity for designing how students should interact with AI. As previously mentioned, AI tutoring styles is not static [3]. The characteristics of the AI's interaction can be engineered using prompts (prompt engineering) and guardrails [14]. In other words, poor learning outcomes are not solely based on the AI's accuracy and performance, but also an issue with the interaction design itself [15, 16].

To address the illusion of truth, AI tools must be designed with guardrails that encourage cognitive engagement rather than handing out direct answers [17]. In the context of learning, the design of the AI interaction itself is more important than its capabilities [16]. Evidence shows that the use of scaffolding techniques, such as guiding hints and follow-up questions, are able to significantly improve students' learning outcomes [16]. AI chatbots with these guardrails are referred to as Socratic chatbots, which are designed to prevent giving out direct answers to encourage self-reflection and deep understanding of the topic [18, 19, 20]. When compared to default GenAI models, Socratic chatbots are able to produce 79% more learning improvements than the default models (127% vs. 48%) during practice assessment [16]. Additionally, unlike the default model where students experience learning loss when AI is removed, students did not experience any learning loss when the Socratic chatbot is removed [16]. This indicates that the default AI interaction may have short-term benefits, but confidence outpaces actual understanding, creating an illusion of competence [16]. Although the Socratic tutoring style works in many contexts, it is not universally superior [21]. In the context of coding, a study comparing the effects of Socratic AI versus normal AI shows that Socratic AI produces a higher error ratio and has no evidence of improving long-term outcomes [21]. Furthermore, learners with lower AI literacy tend to find scaffolding bots frustrating, which further amplifies the need for alternative tutoring styles [18]. This underscores the need to look beyond Socratic tutoring. It is important to explore other tutoring styles that could effectively improve both learning and confidence calibration.

Yet most literature rarely compares alternative interaction designs within the same study. Despite growing evidence on AI-supported learning, most studies evaluate one tutoring style at a time. As a result, the field still lacks clear comparative evidence on which tutoring style best results in balanced learning outcomes as well as psychological learning benefits. In practice, GenAI can be deployed with a spectrum of tutoring styles using prompt engineering and guardrails [14]. Beyond Socratic approaches, evidence shows that reverse tutoring can be effective in boosting cognitive engagement and knowledge mastery [22, 23, 15, 24]. This pedagogical approach embodies the theory of learning-by-teaching by using recursive feedback that observes how a tutee applies what has been taught [22]. This

approach is similar to designs where the AI takes the role of a student, prompting the human learner to teach the AI, thereby reinforcing the learner's own understanding [13]. Studies show that reverse tutoring creates the protégé effect where students learn harder when they know they must teach [22, 24]. There are two mechanisms that align with reverse tutoring [23]:

1. **Generative processing:** Students are encouraged to inherently organize and integrate their knowledge.
2. **Metacognitive monitoring:** Students are able to detect explanation gaps, which trigger self-correction.

Moreover, this socio-psychological interaction shifts the learners mindset to dig deeper into the topic, creating a more flexible mastery of the topic [23, 15].

Hence, in addition to the Socratic tutoring style, this study introduces the reverse tutoring technique. This study compares the three tutoring styles within a single comparative framework, while assessing both cognitive and psychological outcomes. In particular, reverse tutoring, which is grounded in learning-by-teaching theory, despite showing clear effectiveness in improving learning outcomes, remains under-explored in the GenAI context. Therefore, this study aims to fill that gap by asking how different tutoring styles (direct-answer, Socratic, and reverse tutoring) compare in their effects on learning performance, knowledge acquisition, self-efficacy, and confidence calibration. This combined evaluation is important as prior work shows that students can appear more productive with AI but could lose their learning once the tool is removed [8, 16]. Accordingly, this study aims to determine which tutoring style best supports durable learning rather than short-term productivity.

2 Literature Review

2.1 Generative AI in Education

2.2 Learning strategies: direct, Socratic, and learning-by-teaching

2.3 Confidence calibration and psychological impacts

2.4 Research gap

3 Research Question

Research question: How do direct-answer, Socratic, and reverse-tutoring (protégé effect) generative AI tutoring styles compare in terms of (a) student learning performance and knowledge acquisition, (b) self-efficacy, and (c) confidence calibration?

Hypothesis: Reverse tutoring (student teaches the AI as a naive tutee) will lead to better knowledge mastery, stronger self-efficacy, and better alignment between confidence and actual competence than both the Socratic chatbot and the direct-answer chatbot, because teaching forces learners to organise knowledge and exposes gaps (protégé effect), reducing the illusion of competence while supporting deeper learning.

Optional sub-questions: Which style best supports long-term retention? Do self-efficacy and perceived difficulty differ across conditions?

4 Research Plan

4.1 Methodology

4.2 Data and experiment

4.3 Participants and ethics

4.4 Timeline

5 Conclusion

References

[1] D. Sallai, J. Cardoso-Silva, M. Barreto, F. Panero, G. Berrada, and S. Luxmoore, "Approach generative ai tools proactively or risk bypassing the learning process in higher education," 07 2024.

[2] J. Prather, J. Leinonen, N. Kiesler, J. Gorson Benario, S. Lau, S. MacNeil, N. Norouzi, S. Opel, V. Pettit, L. Porter, B. N. Reeves, J. Savelka, I. Smith, David H., S. Strickroth, and D. Zingaro, "Beyond the hype: A comprehensive review of current trends in generative ai research, teaching practices, and tools," in *2024 Working Group Reports on Innovation and Technology in Computer Science Education*, ITiCSE 2024, (New York, NY, USA), p. 300–338, Association for Computing Machinery, 2025.

[3] W. Ma, W. Ma, Y. Hu, and X. Bi, "The who, why, and how of ai-based chatbots for learning and teaching in higher education: A systematic review," *Education and Information Technologies*, vol. 30, pp. 7781 – 7805, 2024.

[4] T. Xu and Y. Chen, "Integrating generative ai with human-centered pedagogy: An innovative path for vocational education," in *Proceedings of the International Conference on Implementing Generative AI into Telecommunication and Digital Innovation 2025*, GAITDI '25, (New York, NY, USA), p. 278–283, Association for Computing Machinery, 2026.

[5] Y. Qian, "Pedagogical applications of generative ai in higher education: A systematic review of the field," *TechTrends*, vol. 69, 06 2025.

[6] Y. Ma, Y. Su, M. Li, Y. Zhang, W. Chai, X. Zhao, and A. Huang, "Preparing students for an ai-driven world: Generative ai and curriculum reform in higher education," *Frontiers of Digital Education*, vol. 2, pp. 1–22, 12 2025.

[7] Q. Xia, Q. Liu, A. Tlili, and T. K. Chiu, "A systematic literature review on designing self-regulated learning using generative artificial intelligence and its future research directions," *Computers & Education*, vol. 240, p. 105465, 09 2025.

[8] D. Wang, S. Yu, X. Jing, Q. Liu, and Q. Lu, "Catalyst for future education: An empirical study on the impact of artificial intelligence generated content on college students' innovation ability and autonomous learning," *Education and Information Technologies*, vol. 30, pp. 9949–9968, 12 2024.

[9] J. Loughlin, "Navigating the artificial intelligence landscape in higher education: A human-centered approach to integrating technology for enhanced learning," 01 2026.

[10] J. Kim, S.-S. Lee, R. Detrick, J. Wang, and N. Li, "Students-generative ai interaction patterns and its impact on academic writing," *STEMPS Faculty Publications*, no. 390, 2025.

[11] Y. Klimova and J. Pikhart, "Over-reliance on ai: A systemic review of its impact on students' learning and development," *Journal of Educational Technology & Society*, vol. 10, pp. 1–12, 01 2025.

[12] L. S. Lo, "The clear path: A framework for enhancing information literacy through prompt engineering," *Journal of Academic Librarianship*, vol. 4, 2023.

[13] E. R. Mollick and L. Mollick, "Assigning ai: Seven approaches for students, with prompts," tech. rep., The Wharton School Research Paper, 09 2023. Available at SSRN: `https://ssrn.com/abstract=4475995` or `http://dx.doi.org/10.2139/ssrn.4475995`.

[14] A. Yazici, P. Mejia-Domenzain, J. Frej, and T. Käser, "Gelex: Generative ai-hybrid system for example-based learning," in *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems*, CHI EA '24, (New York, NY, USA), Association for Computing Machinery, 2024.

[15] B. Lyu, C. Li, H. Li, H. Oh, Y. Song, W. Zhu, and W. Xing, "The role of teachable agents' personality traits on student-ai interactions and math learning," *Computers & Education*, vol. 234, p. 105314, 04 2025.

[16] H. Bastani, O. Bastani, A. Sungu, H. Ge, O. Kabakci, and R. Mariman, "Generative ai without guardrails can harm learning: Evidence from high school mathematics," *Proceedings of the National Academy of Sciences*, vol. 122, no. 26, p. e2422633122, 2025.

[17] H. Wang, Z. Wang, and T. Zhou, *Generative AI and Self-Regulated Learning in Education: A Scientometric Review*, p. 301–307. New York, NY, USA: Association for Computing Machinery, 2026.

[18] Y. Jin, K. Yang, L. Yan, V. Echeverria, L. Zhao, R. Alfredo, M. Milesi, J. X. Fan, X. Li, D. Gasevic, and R. Martinez-Maldonado, "Chatting with a learning analytics dashboard: The role of generative ai literacy on learner interaction with conventional and scaffolding chatbots," in *Proceedings of the 15th International Learning Analytics and Knowledge Conference*, LAK '25, (New York, NY, USA), p. 579–590, Association for Computing Machinery, 2025.

[19] I. Hilliger, M. Pérez-Sanagustín, E. Villalobos, R. Ferreira, and C. Gonzalez, "Scaffolding learning scenarios with a socratic chatbot: Insights from educators," in *Two Decades of TEL. From Lessons Learnt to Challenges Ahead: 20th European Conference on Technology Enhanced Learning, EC-TEL 2025, Newcastle upon Tyne and Durham, UK, September 15–19, 2025, Proceedings, Part II*, (Berlin, Heidelberg), pp. 155–161, Springer-Verlag, 2025.

[20] J. Jabbour, K. Kleinbard, O. Miller, R. Haussman, and V. J. Reddi, "Socratiq: A generative ai-powered learning companion for personalized education and broader accessibility," in *Proceedings of the Workshop on Computer Architecture Education*, WCAE '25, (New York, NY, USA), Association for Computing Machinery, 2025.

[21] Z. Liding, M. Osmolovskiy, H. Lanka, R. Howard, N. Roy, and R. Borela, "To tell or to ask? comparing the effects of targeted vs. socratic ai hints," pp. 1423–1424, 02 2026.

[22] S. Y. Okita and D. L. Schwartz, "Learning by teaching human pupils and teachable agents: The importance of recursive feedback," *Journal of the Learning Sciences*, vol. 22, no. 3, pp. 375–412, 2013.

[23] L. Fiorella, "Learning by teaching," in *In Their Own Words: What Scholars and Teachers Want You To Know About Why and How To Apply the Science of Learning in Your Academic Setting* (C. E. Overson, C. M. Hakala, L. L. Kordonowy, and V. A. Benassi, eds.), pp. 158–170, Society for the Teaching of Psychology, 2023.

[24] S. Wong and S. Lim, "The powerful teacher: A power hypothesis for the benefits of learning-by-teaching," *Educational Psychology Review*, vol. 38, p. 26, 2026.